

Abhishek Singh

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EDUCATION

- Rochester Institute of Technology (RIT)** Rochester, NY
Master of Science in Data Science; GPA: 3.9 Aug. 2023 – May 2026

RESEARCH & DEVELOPMENT EXPERIENCE

- RIT Evolutionary Computing and ML Lab** Rochester, NY
Student Graduate Research Engineer & Data Analyst Jan 2025 – Present
 - Experimental Design:** Engineered an automated experimentation pipeline on a 2,304-core HPC cluster to evolve and test 10,000+ neural architectures for renewable energy time-series (Wind/Solar).
 - Ablation Studies & Optimization:** Currently executing large-scale ablation experiments in C++ to isolate the performance impact of asynchronous selection mechanisms versus structural pruning; analyzing trial data to optimize algorithms for low-power "edge" solar devices.
 - Data Integrity & Analysis:** Processed multi-GB datasets using **Excel** and Python; ensured accuracy of experimental logs and model metadata for reproducible R&D reporting.
 - Collaborative Research:** Documented findings in technical notes and co-authored a peer-reviewed publication; collaborated with senior faculty and mentored undergraduate engineering interns on lab protocols.
- Morningstar** Mumbai, India
Associate Software Engineer May 2021 – May 2023
 - Technical Implementation:** Developed portfolio analytics tools using Vue.js, ensuring high precision in data visualization for complex financial reporting.
 - Process Documentation:** Automated technical documentation workflows via CMS-driven systems, reducing reporting errors and improving team transparency.

PUBLICATIONS

Singh, A., Lyu, Z., Desell, T. "Biologically-Inspired Homeostasis for Neuroevolution: Alternating Growth and Pruning Phases." Accepted at *EvoStar 2026*.

Key Result: 23% reduction in model parameters ($p < 0.05$) while maintaining forecast accuracy.

TECHNICAL PROJECTS

- Generative AI Video Compliance System** Azure, LangGraph
Lead Developer
 - Multi-Modal Testing:** Designed a QA pipeline utilizing Azure Video Indexer and OCR to ingest and analyze regulatory documentation against live video data.
 - Reporting & Metrics:** Implemented LangSmith observability to track system latency and accuracy, presenting data-driven optimization reports to project stakeholders.
- AI Travel Itinerary Planner** GCP, Kubernetes
System Architect
 - Hardware/Software Maintenance:** Operated and maintained containerized microservices on GCP; troubleshooted deployment bottlenecks and ensured 99.9% system availability.

TECHNICAL SKILLS

- Data Analysis:** Microsoft Excel (**Advanced**), Python (Pandas, NumPy), Mann-Whitney U Testing, Mann-Kendall Trend Analysis, Matplotlib/Seaborn Visualization
- Engineering Tools:** Hands-on Lab Testing, C++, Bash Scripting, Linux, Git, Slurm HPC, **Equipment Troubleshooting**, Pydantic (Data Validation)
- AI & Machine Learning:** Distributed Computing, Time-Series Forecasting, Neural Architecture Search, LLMs, RAG, PyTorch, Scikit-learn
- Cloud & Infrastructure:** Azure (AI Search, Monitor), GCP, Docker, Kubernetes, FastAPI, PostgreSQL